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NMAM INSTITUTE OF TECHNOLOGY, NITTE*Off-Campus Centre of Nitte (Deemed to be University)***Second Semester M.Tech (Structural Engineering) (CBCS) Degree Examinations**

May - June 2024

22CST223 – DESIGN OF EARTHQUAKE RESISTANT STRUCTURES

Duration: 3 Hours

Max. Marks: 100

Note: Answer Five full questions choosing Two full questions from Unit – I & Unit – II each and One full question from Unit – III.

❖ Use of IS: 1893:2016 is allowed.

Unit – I		Marks	BT*	CO*	PO*
1. a)	Describe phenomenon of earthquake in detail.	10	L*2	1	1
b)	Discuss these seismic terms: Epicenter, Epicentral Distance, Focus, Focal depth.	10	L2	1	1
2. a)	Discuss Seismic Zone map of India as per IS – 1893 – 2016.	10	L4	2	2
b)	Explain Zone of Divergence, Zone of Convergence and Fracture Zone.	10	L2	1	1
3. a)	Describe about Seismic Design Philosophy.	10	L4	1	1
b)	How to measure earthquake. Discuss in detail.	10	L2	1	1
Unit – II					
4.	Discuss key Points of IS 1893:2016 Part 1.	20	L4	2	2
5.	For a four storeyed RCC office building located in zone V and resting on hard rock, compute the seismic forces as per IS-1893-2016 equivalent static procedure. Height of first floor is 4.2 m and the remaining three stories are of height 3.5 m each. Plan dimensions (length and width) of the structure are 10 m x 15 m. The RCC frames are infilled with brick masonry. Dead load on floor 15 kN/m ² and 12 kN/m ² on roof. Live = 4 kN/m ² on floors and 1.5 kN/m ² on roof. Also compute the base shear, neglecting the stiffness of infill walls. Compare the base shears for the two cases and comment on the result.	20	L5	2	2
6.	A Ten-story hospital building is to be constructed at Chennai. The foundation is to be provided is raft over medium soil. Using IS1893-2016 (Part-I) determine the earthquake forces in each story in the structural system consist of OMRF without brick infill panels. Given Data: The LL on each Floor = 6000 N/m ² The LL on Roof = 2500 N/m ² The DL of Floor Finish (Slab and Beam) = 4000 N/m ² Height of each Story=3.2m Spacing of Columns in both the directions = 4.5m	20	L5	3	3
Unit – III					
7.	Explain Earthquake effect on masonry buildings in detail.	20	L2	4	4
8.	Brief about earthquake resistant design of masonry building.	10	L2	5	5

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
